Smooth isometric embeddings of closed orientable surfaces in four-space

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1 Problem and notation

All manifolds, maps, and metrics are smooth, meaning C^∞. A closed manifold is compact and has no boundary. Write δ_N for the Euclidean metric on $\mathbb{R}^N$ and $\gamma(F) = F^*\delta_N$ for the metric induced by an immersion. Fix a smooth background metric g_{bg} and a finite atlas when taking norms. All $o(1)$ estimates below are uniform on the specified closed sets.

Theorem 1.1 (Smooth isometric embedding)**.** *Every closed orientable smooth surface M with any smooth Riemannian metric g admits a smooth embedding $f : M \hookrightarrow \mathbb{R}^4$ satisfying*

$$\langle \mathrm{d}f_p(v), \mathrm{d}f_p(w)\rangle = g_p(v,w) \qquad (p \in M,\ v, w \in T_pM). \tag{1.1}$$

No curvature condition is imposed.

Equation (1.1) expresses preservation of the Riemannian metric on tangent spaces. The proof uses exact primitive realization, boundary propagation, and uniform spherical preparation in the forms stated below. We show that these operations preserve global injectivity.

2 Primitive metric corrections

We use exact primitive addition with the following local geometric data and boundary conditions. Our convention for the vector-valued second fundamental form is

$$F_{ij} = \Gamma_{ij}^k F_k + B_F(\partial_i, \partial_j). \tag{2.1}$$

Definition 2.1 (Admissible primitive data)**.** Let $F : M \to \mathbb{R}^4$ be an immersion of a closed surface, with B_F nowhere the zero tensor and with a smooth global unit normal n. Let D be an open coordinate disk with smooth boundary C_0, and let (x, y) be coordinates on a neighborhood of $\overline{D}$. Take a smooth function $a \geq 0$ on M, positive exactly on D, and put $Y_0 = \partial_y$. Require

$$\mathrm{Hess}_{\gamma(F)}\, x(\partial_y, \partial_y) > 0 \quad \text{on } \overline{D}. \tag{2.2}$$

Let $C_1, \ldots, C_m$ be later smooth embedded closed curves, with smooth nonzero vector fields Y_i on their neighborhoods. The fields need not be tangent to the curves. Require:

(i) The curves $C_0, \ldots, C_m$ are pairwise transverse and have no triple intersection.

(ii) On $C_i \cap \overline{D}$, for $i > 0$, the field ∂_y is tangent to C_i at only finitely many points.

(iii) At $C_i \cap C_l \cap D$, for distinct $i, l > 0$, both $\mathrm{d}x(Y_i)$ and $\mathrm{d}x(Y_l)$ are nonzero.

(iv) With $A_i = B_F(Y_i, Y_i)$ and

$$Q_{il} = \left\langle B_F(Y_i, Y_l), \frac{A_i}{|A_i|} \right\rangle^2 + K_{\gamma(F)} |Y_i \wedge Y_l|^2_{\gamma(F)}, \tag{2.3}$$

the following conditions hold for all indices $0, \dots, m$:

$$\begin{aligned} A_i \neq 0, \qquad & A_i/|A_i| \neq -n \quad \text{on } C_i, \\ \langle A_i, n \rangle > 0, \qquad & Q_{il} > 0 \qquad\quad \text{on } C_i \cap C_l \quad (i \neq l). \end{aligned} \tag{2.4}$$

The crossing conditions are imposed in both orders.

2.1 Exact realization and uniform asymptotics

For these data, the velocity-loop construction uses the plane, projected velocity, and radius

$$P_p = \operatorname{span}(F_y(p), F_{yy}(p))^\perp, \qquad v = \pi_P F_x, \qquad R_0 = (|v|^2 + a^2)^{1/2}. \tag{2.5}$$

There is a smooth 2π-periodic velocity $V(p, \theta)$, also smooth in the two-jet of a nearby map, such that

$$V \in P, \qquad |V| = R_0 > 0, \qquad \operatorname{avg}_\theta V = v. \tag{2.6}$$

The difference $V - v$ extends smoothly by zero across ∂D. All loop choices, including the finite turn threshold used to preserve later boundary data, are fixed before choosing the small scale z.

The exact primitive-realization theorem supplies, for every sufficiently small $z > 0$, a smooth immersion f_z such that

$$\gamma(f_z) = h_* := \gamma(F) + a^2 \mathrm{d}x^2. \tag{2.7}$$

The primitive tensor is extended by zero. With $X = F_x - v + V$, the same theorem gives the uniform estimates on $\overline{D}$

$$\begin{aligned} (f_z)_x &= F_x - v + V(p, x/z) + o(1), & (f_z)_y &= F_y + o(1), \\ (f_z)_{yy} &= F_{yy} + o(1), & (f_z)_{xy} &= X_y + o(1), \\ z(f_z)_{xx} &= V_\theta + o(1). \end{aligned} \tag{2.8}$$

Periodic arguments are evaluated at $\theta = x/z$; X_y holds θ fixed. On $M \setminus D$, one has $f_z \to F$ in C^2.

These estimates hold for the *final exact maps*: the final correction is $o(1)$ in C^2 and retains the displayed asymptotics. The realization theorem also supplies a uniform positive lower bound for the size of the second fundamental form.

2.2 The same maps preserve the later conditions

The boundary-propagation theorem uses the same exact maps with the loop threshold fixed first. For all sufficiently small z, it supplies a global preferred unit normal n_z, preserves the nowhere-zero second-form tensor, and establishes (2.4) for all later indices. The adjustments at crossings and at the collar are adjustments of normal fields; they do not replace f_z. Consequently an additional condition that holds for all sufficiently small z may be imposed on these same exact maps.

3 A geometric stability lemma

The results in this section do not depend on the analytic construction. Let $F : M \hookrightarrow \mathbb{R}^N$ be an embedding of a closed manifold, and write $T_p = \mathrm{d}F_p(T_pM)$.

Lemma 3.1 (Tubular coordinates for a complement)**.** *Let E be a smooth subbundle of $M\times\mathbb{R}^N$ such that $\mathbb{R}^N = T_p \oplus E_p$ for every p. For some $r > 0$, the map*

$$\Psi : E_{<r} \longrightarrow \mathbb{R}^N, \qquad \Psi(p, e) = F(p) + e, \qquad E_{<r} = \{(p, e) \in E : |e| < r\},$$

is a diffeomorphism onto an open neighborhood U of $F(M)$. The retraction $H = F \circ \pi \circ \Psi^{-1} : U \to F(M)$ satisfies

$$DH_{F(p)}|_{E_p} = 0, \qquad DH_{F(p)}\mathrm{d}F_p = \mathrm{d}F_p. \tag{3.1}$$

Proof. At the zero section, the natural tangent splitting gives $D\Psi_{(p,0)}(v, e) = \mathrm{d}F_p(v) + e$, an isomorphism. The inverse function theorem and compactness imply that Ψ is a local diffeomorphism on $E_{<r}$ for a sufficiently small uniform r.

If no smaller radius gave injectivity, there would be unequal pairs $(p_j, e_j), (q_j, e'_j)$ with $|e_j|, |e'_j| \to 0$ and $F(p_j) + e_j = F(q_j) + e'_j$. After taking subsequences, $p_j \to p$ and $q_j \to q$. Then $F(p) = F(q)$, so $p = q$. Both pairs converge in the total space to $(p, 0)$, contradicting local injectivity there. Thus Ψ is an injective local diffeomorphism after shrinking r. Its image is open and its inverse smooth. The identities follow by differentiating $H(F(p)+e) = F(p)$. □

Lemma 3.2 (C^1 openness of compact embeddings)**.** *If smooth maps $G_j : M \to \mathbb{R}^N$ converge to F in C^1, then G_j is an embedding for all sufficiently large j.*

Proof. Near any point choose coordinates u and a linear map $L : \mathbb{R}^N \to \mathbb{R}^{\dim M}$ so that $D(LF)$ at the center is the identity. On a sufficiently small convex coordinate ball, its derivative differs from the identity by less than $1/4$. For all sufficiently large j, $D(LG_j)$ differs from the identity by less than $1/2$ there. Integrating the derivative error along a segment yields

$$|LG_j(u) - LG_j(v)| \geq \tfrac{1}{2}|u - v|.$$

This proves injectivity and immersion on that ball. Take finitely many such balls whose smaller concentric balls cover M. The union of the smaller-ball products is a neighborhood of the diagonal in $M \times M$. Its complement is compact and disjoint from the diagonal. If nonempty, $|F(p) - F(q)|$ has a positive minimum there; uniform convergence preserves this separation. Thus G_j is an injective immersion on all of M. A continuous injection from compact M into a Hausdorff space is a homeomorphism onto its image. □

Proposition 3.3 (Embedding preservation under transverse oscillations)**.** *Let E be as in Lemma 3.1. Suppose smooth maps $f_j : M \to \mathbb{R}^N$ satisfy*

$$f_j \to F \text{ in } C^0, \qquad \sup_j \|\mathrm{d}f_j\|_{C^0} < \infty, \qquad \mathrm{d}f_j - \mathrm{d}F = W_j + R_j, \tag{3.2}$$

where $W_{j,p}(T_pM) \subset E_p$ and $\|R_j\|_{C^0} \to 0$. Then f_j is an embedding for all sufficiently large j.

Proof. For large j, uniform convergence puts $f_j(M)$ in the fixed tubular neighborhood U. At each p, the chain rule gives

$$\begin{aligned} \mathrm{d}(H \circ f_j)_p - \mathrm{d}F_p &= (DH_{f_j(p)} - DH_{F(p)})\mathrm{d}f_{j,p} + DH_{F(p)}(\mathrm{d}f_{j,p} - \mathrm{d}F_p) \\ &= (DH_{f_j(p)} - DH_{F(p)})\mathrm{d}f_{j,p} + DH_{F(p)}R_{j,p}. \end{aligned} \tag{3.3}$$

The term W_j vanishes exactly under $DH_{F(p)}$. On a compact subneighborhood of $F(M)$, DH is uniformly continuous. The first term therefore tends uniformly to zero by the boundedness of $\mathrm{d}f_j$, and the second does so by the assumption on R_j. Also $H \circ f_j \to F$ uniformly, so the convergence is in C^1. Lemma 3.2 makes $H \circ f_j$ an embedding for large j.

If $f_j(p) = f_j(q)$, injectivity of the composition gives $p = q$. If $\mathrm{d}f_{j,p}(v) = 0$, injectivity of $\mathrm{d}(H \circ f_j)_p$ gives $v = 0$. Thus f_j is an injective immersion, and compactness finishes the proof. □

Remark 3.4. The same argument applies to a family indexed by $z \downarrow 0$, with no continuity in z required. The composition $H \circ f_z$ is only a test for injectivity; it does not replace the exact isometric map f_z. No metric identity is asserted for the composition.

4 Embedding preservation in one primitive addition

Lemma 4.1 (Transversality of the oscillation plane)**.** *Under* (2.2)*, the planes P in* (2.5) *form a smooth two-plane bundle near $\overline{D}$, and*

$$\mathbb{R}^4 = \mathrm{d}F_p(T_pM) \oplus P_p. \tag{4.1}$$

Proof. Since x is a coordinate function, $\mathrm{Hess}_{\gamma(F)}\, x(\partial_y, \partial_y) = -\Gamma^x_{yy}$. Thus $\Gamma^x_{yy} \neq 0$. By (2.1),

$$F_{yy} = \Gamma^x_{yy} F_x + \Gamma^y_{yy} F_y + B_{yy}.$$

The tangential projections of F_y and F_{yy} are independent, so the ambient vectors are independent and their perpendicular complement is a smooth two-plane bundle. If $w \in P_p \cap \mathrm{d}F_p(T_pM)$, then $\langle w, F_y\rangle = \langle w, F_{yy}\rangle = 0$ and $\langle w, B_{yy}\rangle = 0$. Hence $\Gamma^x_{yy}\langle w, F_x\rangle = 0$, which implies $w = 0$. Strict positivity on the compact set $\overline{D}$ persists on a neighborhood, so the assertions hold there as well. □

The associated determinant is

$$\det \begin{pmatrix} \langle F_y, F_x\rangle & \langle F_y, F_y\rangle \\ \langle F_{yy}, F_x\rangle & \langle F_{yy}, F_y\rangle \end{pmatrix} = -\Gamma^x_{yy} \det \gamma(F) > 0. \tag{4.2}$$

This transversality will let us remove the leading oscillation by projection along a complement to the old tangent bundle.

Remark 4.2 (Nonvanishing of the velocity radius). The Hessian condition alone does not imply $v \neq 0$ everywhere. In D, $a > 0$ gives $|v|^2 + a^2 > 0$. On ∂D, the boundary condition gives $B_{yy} \neq 0$ and projection yields $\pi_P B_{yy} = -\Gamma^x_{yy} v$. Projection from the normal plane to P is an isomorphism: its kernel would lie in $\mathrm{span}(F_y, F_{yy})$ with zero tangential projection, and that projection is injective. Hence $v \neq 0$ at the boundary and on a collar. Compactness supplies a uniform positive radius margin.

Lemma 4.3 (Extension of a local complement)**.** *If F is an embedding, there is a smooth global complement E to $\mathrm{d}F(TM)$ agreeing with P near $\overline{D}$.*

Proof. Write $T = \mathrm{d}F(TM)$ and $N = T^\perp$. On the neighborhood where P is defined, $\pi_N|_P : P \to N$ is a bundle isomorphism. Thus $P_p = \{A_p u + u : u \in N_p\}$ for a smooth bundle map $A : N \to T$. Multiply A by a cutoff equal to one near $\overline{D}$ and supported in that neighborhood, and extend the product by zero to a global smooth bundle map $\overline{A}$. Its graph is the required E. Every such graph is transverse to T, regardless of its norm. □

Lemma 4.4 (Uniform displacement)**.** *The exact maps in (2.8) converge to F in $C^0(M)$.*

Proof. Set $e_z = f_z - F$. Exterior convergence controls e_z uniformly on ∂D, while (2.8) gives $\|\partial_y e_z\|_{C^0(\overline{D})} \to 0$. For a point $(x_0, y_0) \in D$, take the connected interval containing y_0 in the vertical slice $\{y : (x_0, y) \in D\}$. It has an endpoint on ∂D and length at most the finite y-range L_D of the coordinate image of $\overline{D}$. Integrating to that endpoint gives

$$\|e_z\|_{C^0(\overline{D})} \le \|e_z\|_{C^0(\partial D)} + L_D \|\partial_y e_z\|_{C^0(\overline{D})} \longrightarrow 0. \tag{4.3}$$

Combine this with exterior convergence. Vertical convexity of the whole disk and continuous choices of endpoints are unnecessary. □

Proposition 4.5 (Embedding-preserving primitive addition)**.** *For admissible primitive data, suppose that F is an embedding. The exact primitive output F_+ can be chosen to be an embedding, retaining its target metric, nowhere-zero second-form tensor, preferred normal, and later boundary conditions. Its exterior C^2 error and global C^0 displacement can be made arbitrarily small.*

Proof. Fix the velocity loop, and let f_z be its exact realizations. For the global complement from Lemma 4.3, set

$$W_z = (V(p, x/z) - v(p))\,\mathrm{d}x \quad \text{on } D, \qquad W_z = 0 \quad \text{outside } D.$$

For each fixed z, this is a smooth global one-form with values in E, by the smooth zero extension of $V - v$. The first-derivative asymptotics and exterior convergence give

$$\mathrm{d}f_z - \mathrm{d}F = W_z + R_z, \qquad \|R_z\|_{C^0(M)} \to 0.$$

For background-unit u, the exact metric identity gives $|\mathrm{d}f_z(u)|^2 = h_*(u, u)$. Since h_* is fixed, the differentials are uniformly bounded. Lemma 4.4 gives the remaining hypothesis of Proposition 3.3, so f_z is an embedding for all sufficiently small z.

By the boundary-propagation theorem in Section 2, the other geometric conclusions hold for these same maps for every sufficiently small z. Choose z below all the thresholds, also meeting the requested exterior error and displacement, and set $F_+ = f_z$. □

5 Embedded preparation and finite assembly

Lemma 5.1 (Embedded spherical preparation)**.** *For a closed connected orientable surface, uniform spherical preparation and global metric preparation can be performed with the initial map F^0 an embedding, retaining the required geometric conditions and prescribed intermediate metrics.*

Proof. A closed connected orientable smooth surface is diffeomorphic to the boundary of a handlebody in $\mathbb{R}^3$. Compose such an embedding with inverse stereographic projection to obtain $I : M \hookrightarrow \mathbb{S}^3$. No metric is prescribed for this starting embedding.

A uniform family inside the embedding neighborhood. The uniform spherical-preparation theorem applies to this fixed I and every finite set $P \subset M$. It supplies constants $C, c, \varepsilon_0 > 0$ such that, for any $0 < \varepsilon < \varepsilon_0$, one can choose $I_P : M \to \mathbb{S}^3$ with

$$\begin{aligned} B^{\mathbb{S}^3}_{I_P}|_P = 0, \qquad & \|I_P - I\|_{C^1} < \varepsilon, \\ \|I_P\|_{C^2} \le C, \qquad & I_P^* \delta_4 \ge c\, g_{\mathrm{bg}}. \end{aligned} \tag{5.1}$$

The constants are independent of the number and separation of points of P. Fix ε small enough for Lemma 3.2. Then every permitted I_P is an embedding, including maps whose finite set is chosen later. Restrict the preparation family to this neighborhood.

The order of choices in the global construction. Global metric preparation begins with a fixed spherical immersion and its uniform preparation family. Use the embedding I just chosen. First choose $\lambda > 0$ uniformly for the restricted family so that

$$\gamma^0_{\overline{I}} = \lambda^2 \overline{I}^* \delta_4 < g, \qquad h_{\overline{I}} = g - \gamma^0_{\overline{I}} > 0.$$

The bounds in (5.1) give uniform positive lower bounds and C^1 bounds for all path metrics $\gamma^0_{\overline{I}} + s h_{\overline{I}}$, $0 \le s \le 1$. In particular their Christoffel symbols are uniformly bounded.

Uniform metric preparation gives convex phases, disks, and allowed perturbation sizes for this family, with strict cone and phase-Hessian margins. Choose a number N_c of cycles sufficiently large to retain these margins at every partial metric. Each perturbed cycle has an invertible bundle endomorphism $\mathcal{A}_c$ of symmetric tensors and amplitudes

$$a_j = N_c^{-1/2} \psi_j \sqrt{q^0_{\nu r}(\mathcal{A}_c^{-1} h_{\overline{I}})}, \qquad \sum_{j \text{ in cycle } c} a_j^2 \mathrm{d}x_j^2 = N_c^{-1} h_{\overline{I}}. \tag{5.2}$$

Here $q^0_{\nu r}$ are the coefficient functionals of its positive rank-one decomposition, and ψ_j are its support cutoffs. These formulas and their uniform bounds are part of the metric-preparation theorem. Only the already fixed bounds on the family enter these choices.

Now choose the allowed generic boundary and phase perturbations. Their boundary-crossing set P is finite. Choose I_P only at this stage and set $F^0 = \lambda I_P$, $n^0 = -I_P$. Insert the actual deficit $h = g - \gamma(F^0)$ in (5.2). The uniform preparation estimates retain the cone, Hessian, and exact-cycle identities for this choice. Since I_P is an embedding, so is F^0. No earlier constant needs to be chosen again after selecting P.

Initial boundary geometry. For $\gamma^0 = \gamma(F^0)$, the radial decomposition is

$$B_{F^0}(v, w) = \lambda B^{\mathbb{S}^3}_{I_P}(v, w) + \lambda^{-1} \gamma^0(v, w) n^0. \tag{5.3}$$

Its dot product with n^0 is positive for $v = w \neq 0$, so the second-form tensor is nowhere zero and every pure direction avoids $-n^0$. At a point of P, the spherical term vanishes. The Gauss equation gives $K_{\gamma^0} = \lambda^{-2}$ and

$$\begin{aligned} Q_{il}(F^0) &= \lambda^{-2}(\gamma^0(Y_i, Y_l)^2 + |Y_i \wedge Y_l|^2_{\gamma^0}) \\ &= \lambda^{-2} \gamma^0(Y_i, Y_i) \gamma^0(Y_l, Y_l) > 0. \end{aligned} \tag{5.4}$$

This verifies both ordered crossing conditions. All initial hypotheses of the finite construction are therefore retained. □

Proof of Theorem 1.1. Assume first that M is connected. Lemma 5.1 gives an embedded F^0 and a finite list (D_j, x_j, y_j, a_j), $1 \le j \le K$, with exact prescribed metrics

$$\begin{gathered} \gamma^j = \gamma(F^0) + \sum_{i=1}^{j} a_i^2 \mathrm{d}x_i^2, \qquad \gamma^K = g, \\ \mathrm{Hess}_{\gamma^{j-1}} x_j(\partial_{y_j}, \partial_{y_j}) > 0 \quad \text{on } \overline{D}_j. \end{gathered} \tag{5.5}$$

Suppose F^{j-1} is an embedding with induced metric γ^{j-1}, nowhere-zero second-form tensor, and a preferred normal satisfying all remaining boundary conditions. The geometric curve

conditions are fixed by preparation; the Hessian condition is intrinsic and follows from (5.5). Proposition 4.5 gives an embedding F^j inducing γ^j and retaining the conditions needed for the next step. This closes the finite induction. Its final map $f = F^K$ is an embedding with induced metric g.

Each scale is chosen after fixing the current smooth map. Constants and tubular radii need not be uniform between stages. At each stage, Proposition 4.5 applies to the exact output of the primitive-realization theorem. No injectivity assertion is inferred from a mere C^0 limit of embeddings.

A compact manifold has finitely many connected components because its components are open. Apply the connected construction to each component and translate the finitely many compact images to make them disjoint. Translations preserve the induced metric. The empty case is vacuous. □

6 Conclusion

The construction starts from an embedded surface in a small round three-sphere and decomposes the prescribed metric change into finitely many primitive additions. At each step, the positive transverse phase Hessian makes the oscillation plane complementary to the old tangent plane. An oblique tubular retraction cancels the leading derivative oscillation, so its composition with the exact corrected map is C^1 close to the previous embedding. This proves injectivity of the corrected map itself. The corrected map retains the exact new metric and the geometric conditions needed for the next addition.

Uniform spherical preparation provides the initial embedding and the finite metric sequence. Applying the embedding-preserving primitive step along this sequence gives a final smooth embedding with the prescribed metric. Thus, using the exact primitive-realization, boundary-propagation, and preparation theorems in the forms stated above, we conclude that every closed orientable smooth Riemannian surface (M, g) admits a smooth embedding

$$\boxed{f : M \hookrightarrow \mathbb{R}^4, \qquad f^*\delta_4 = g.}$$

The metric g is arbitrary; no curvature restriction is required. For disconnected surfaces, translations separate the finitely many component images while preserving their induced metrics.